

Factsheet for the finite implication $E677 \models_{\text{fin}} E255$

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Abstract

Checked local facts for finite magmas satisfying

$$E677 : \quad x = y \diamond (x \diamond ((y \diamond x) \diamond y)), \quad E255 : \quad x = ((x \diamond x) \diamond x) \diamond x.$$

Fix x . Put $a = x \diamond x$, $q = a \diamond x$, and $p = x \backslash x$. The target is

$$E255 \text{ at } x \iff q \diamond x = x.$$

Everything below is proved here from E677 plus finiteness, or marked as external ETP input. No right cancellation, associativity, identity element, or quasigroup structure is used.

1 Setup and finite tools

A magma is a set with one binary operation $\diamond$. For $u \in M$, write

$$L_u(t) = u \diamond t, \quad R_u(t) = t \diamond u, \quad S(t) = t \diamond t.$$

If L_u is bijective, $u \backslash v$ denotes the unique w with $u \diamond w = v$. All indices on an L_x -cycle are read modulo its period.

Proposition 1 (Finite left division). *In every finite E677-magma, each L_a is bijective and*

$$a \backslash b = b \diamond ((a \diamond b) \diamond a).$$

Also

$$a \backslash (a \backslash b) = (a \backslash b) \diamond (b \diamond a), \quad b \backslash (a \backslash b) = (a \diamond b) \diamond a.$$

Proof. Proof certificate. E677 gives

$$b = a \diamond (b \diamond ((a \diamond b) \diamond a)),$$

so L_a is surjective, hence bijective by finiteness. Substitute $a \backslash b$ for b , and use uniqueness of left division. $\square$

Lemma 2 (Two global identities). *For all y, z ,*

$$z = (y \diamond z) \diamond ((y \diamond (y \diamond z)) \diamond y) \quad (\text{T})$$

and

$$(y \diamond (y \diamond z)) \diamond y = z \diamond (((y \diamond z) \diamond z) \diamond (y \diamond z)). \quad (\text{K})$$

Proof. Proof certificate. For (T), substitute $y \diamond z$ for the law-variable x in E677, then left-cancel y . For (K), compare (T) with E677 using parameter $y \diamond z$, then left-cancel $y \diamond z$. $\square$

Remark 3 (External ETP guardrails). The unrestricted implication $\text{E677} \models \text{E255}$ is false in the free E677-magma, so any proof must use finiteness. Linear models satisfying E677 satisfy E255; affine-fiber extensions over them are ruled out by ETP structure theory. Finite E677-magmas need not be right-cancellative [3].

2 Local equivalences

Fix a finite E677-magma M and $x \in M$. Put

$$a = x \diamond x, \quad q = a \diamond x = (x \diamond x) \diamond x, \quad p = x \backslash x.$$

Proposition 4 (Unique right fixed-point witness). *If $u \diamond x = x$, then $u = q$. Hence*

$$\text{E255 at } x \iff \exists u (u \diamond x = x) \iff q \diamond x = x.$$

Proof. Proof certificate. If $u \diamond x = x$, then (T) gives $x = x \diamond (x \diamond u)$. The specialization $(x, y) = (x, x)$ in E677 gives $x = x \diamond (x \diamond q)$. Two left cancellations give $u = q$. $\square$

Proposition 5 (Equivalent local targets). *The following are equivalent:*

$$\text{E255 at } x, \quad q \diamond x = x, \quad q \backslash x = x, \quad ((x \diamond x) \diamond q) \diamond (x \diamond x) = x, \quad (q \diamond x) \diamond q = p.$$

The fourth identity is the identity sometimes called Lemma L.

Proof. Proof certificate. The first two are Proposition 4. (T), with $(y, z) = (x \diamond x, x)$, gives

$$x = q \diamond (((x \diamond x) \diamond q) \diamond (x \diamond x)).$$

Thus $q \backslash x = ((x \diamond x) \diamond q) \diamond (x \diamond x)$. Finally

$$q \backslash x = x \diamond ((q \diamond x) \diamond q),$$

and left cancellation by x compares this with $p = x \backslash x$. $\square$

Lemma 6 (Fixed maps). *Define*

$$H_x(t) = ((x \diamond t) \diamond x), \quad F_x(t) = x \diamond (t \diamond x).$$

Then

$$x \setminus t = t \diamond H_x(t), \quad F_x(L_x(t)) = L_x(H_x(t)).$$

Thus F_x and H_x have the same finite cycle structure. Moreover E255 at x is equivalent to either map having a fixed point. If fixed points exist, they are unique:

$$\text{Fix}(F_x) = \{x \setminus q\}, \quad \text{Fix}(H_x) = \{x \setminus (x \setminus q)\}.$$

Under $q \diamond x \neq x$, both maps are fixed-point-free.

Proof. Proof certificate. E677 with parameter x gives

$$t = x \diamond (t \diamond ((x \diamond t) \diamond x)) = x \diamond (t \diamond H_x(t)),$$

hence $x \setminus t = t \diamond H_x(t)$. Also

$$F_x(L_x(t)) = x \diamond ((x \diamond t) \diamond x) = x \diamond H_x(t) = L_x(H_x(t)),$$

so $F_x = L_x H_x L_x^{-1}$, and F_x, H_x are conjugate.

Suppose $F_x(t) = t$. Comparing

$$x \diamond (t \diamond x) = t = x \diamond (t \diamond H_x(t))$$

and left-cancelling first x and then t , we get $H_x(t) = x$. Thus $(x \diamond t) \diamond x = x$, so Proposition 4 gives $x \diamond t = q$; hence $t = x \setminus q$. Conversely, if $q \diamond x = x$ and $t = x \setminus q$, then $H_x(t) = (x \diamond t) \diamond x = q \diamond x = x$, and the displayed edge formula gives

$$t = x \diamond (t \diamond x) = F_x(t).$$

Thus F_x has a fixed point exactly when E255 at x holds, and then its only fixed point is $x \setminus q$. The statement for H_x follows by conjugacy: $H_x(s) = s$ iff $F_x(L_x(s)) = L_x(s)$. Hence, when fixed points exist,

$$L_x(s) = x \setminus q, \quad s = x \setminus (x \setminus q).$$

If $q \diamond x \neq x$, Proposition 4 says that no right fixed point over x exists, so neither F_x nor H_x has a fixed point. $\square$

3 Orbit facts

Define

$$c_0 = x, \quad c_{k+1} = x \diamond c_k,$$

and let d be the minimal period of the L_x -orbit of x . Put

$$d_k = c_k \diamond x.$$

Proposition 7 (Basic orbit package). *For all k ,*

$$c_{k-1} = c_k \diamond d_{k+1}, \quad d_1 = c_{d-2},$$

$$x \backslash c_k = c_{k-1}, \quad c_k \backslash c_{k-1} = d_{k+1}.$$

In particular

$$p = c_{d-1}, \quad q = x \backslash p = d_1 = c_{d-2} = a \diamond x, \quad p \diamond a = q, \quad p \backslash q = a.$$

If $d_j = x$, then $j \equiv d - 2 \pmod{d}$.

Proof. Proof certificate. Applying E677 with parameter x and law-variable c_k gives

$$c_k = x \diamond (c_k \diamond ((x \diamond c_k) \diamond x)) = x \diamond (c_k \diamond d_{k+1}).$$

Since also $c_k = x \diamond c_{k-1}$, left cancellation by x gives

$$c_{k-1} = c_k \diamond d_{k+1}.$$

The equality $x \diamond q = p$ follows from the specialization

$$x = x \diamond (x \diamond q)$$

of E677 at $(x, y) = (x, x)$, because $p = x \backslash x$. Therefore $q = x \backslash p = c_{d-2}$, while $q = a \diamond x = d_1$ by definition. The division statements follow from $x \diamond c_{k-1} = c_k$ and $c_k \diamond d_{k+1} = c_{k-1}$. In particular $p = c_{d-1}$, $p \diamond a = q$ is the recurrence at $k = d - 1$, and $p \backslash q = a$.

Finally suppose $d_j = x$. By (T) with $(y, z) = (c_j, x)$,

$$x = x \diamond ((c_j \diamond x) \diamond c_j) = x \diamond (x \diamond c_j).$$

Comparing with $x = x \diamond (x \diamond q)$ and left-cancelling x twice gives $c_j = q$. Since $q = c_{d-2}$ on the L_x -cycle, $j \equiv d - 2 \pmod{d}$. $\square$

Corollary 8 (Small periods). *If $d = 1$, then E255 at x holds. The cases $d = 2$ and $d = 3$ are impossible. Thus a bad point has $d \geq 4$.*

Proof. Proof certificate. $d = 1$ means $x \diamond x = x$, so $q = x$ and $q \diamond x = x$.

If $d = 2$, then $q = c_{d-2} = c_0 = x$, hence $d_1 = q \diamond x = x$, contradicting the forced-index statement in Proposition 7, since $1 \not\equiv 0 \pmod{2}$.

Assume $d = 3$. Then

$$c_0 = x, \quad c_1 = a = q, \quad c_2 = p,$$

and

$$a \diamond x = a, \quad x \diamond a = p, \quad x \diamond p = x, \quad p \diamond a = a.$$

The orbit recurrence at $k = 1$ gives

$$x = a \diamond (p \diamond x). \tag{D3.1}$$

By (K) with $(y, z) = (x, a)$,

$$a = (x \diamond (x \diamond a)) \diamond x = a \diamond (((x \diamond a) \diamond a) \diamond (x \diamond a)) = a \diamond (a \diamond p).$$

Since also $a \diamond x = a$, left cancellation by a gives

$$a \diamond p = x. \tag{D3.2}$$

Comparing (D3.1) with (D3.2) and left-cancelling a gives

$$p \diamond x = p. \tag{D3.3}$$

Now (K) with $(y, z) = (p, a)$ gives

$$x = (p \diamond (p \diamond a)) \diamond p = a \diamond (((p \diamond a) \diamond a) \diamond (p \diamond a)) = a \diamond ((a \diamond a) \diamond a).$$

Since $a \diamond p = x$, left cancellation by a gives

$$(a \diamond a) \diamond a = p. \tag{D3.4}$$

Finally, (K) with $(y, z) = (a, x)$ gives

$$(a \diamond a) \diamond a = x \diamond (a \diamond a).$$

Using (D3.4) and $x \diamond a = p$, left cancellation by x gives $a \diamond a = a$. Then $(a \diamond a) \diamond a = a$, contradicting (D3.4) and the distinctness of $a = c_1$ and $p = c_2$ on a period-three orbit. $\square$

Lemma 9 (Uniform orbit-edge identities). *For all k ,*

$$d_{k+2} = c_k \diamond ((c_{k+1} \diamond c_k) \diamond c_{k+1}), \tag{U1}$$

$$x = d_k \diamond ((c_k \diamond d_k) \diamond c_k), \tag{U2}$$

$$(c_k \diamond d_k) \diamond c_k = x \diamond ((d_k \diamond x) \diamond d_k). \tag{U3}$$

Therefore

$$d_k \backslash x = (c_k \diamond d_k) \diamond c_k = x \diamond ((d_k \diamond x) \diamond d_k).$$

At a bad point,

$$a \backslash x = d_2, \quad q \backslash x = (a \diamond q) \diamond a = x \diamond ((q \diamond x) \diamond q), \quad q \backslash a = (p \diamond q) \diamond p.$$

Proof. Proof certificate. (U1) is (K) with $(y, z) = (x, c_k)$. (U2) is (T) with $(y, z) = (c_k, x)$. (U3) is (K) with $(y, z) = (c_k, x)$. The quotient forms read (U2) as left division and use (U3).

For the displayed specializations, $a \backslash x = x \diamond ((a \diamond x) \diamond a) = x \diamond (q \diamond a)$, and (U1) at $k = 0$ says $d_2 = x \diamond (q \diamond a)$. Since $d_1 = q$, the general quotient formula at $k = 1$ gives

$$q \backslash x = (a \diamond q) \diamond a = x \diamond ((q \diamond x) \diamond q).$$

Finally $p \diamond a = q$, and (T) with $(y, z) = (p, a)$ gives

$$a = q \diamond ((p \diamond q) \diamond p),$$

so $q \backslash a = (p \diamond q) \diamond p$. $\square$

Proposition 10 (Period-four gate). *Assume $d = 4$. Write*

$$a = c_1, \quad q = c_2, \quad p = c_3, \quad r = q \diamond x, \quad s = p \diamond x.$$

Then

$$x \diamond a = q, \quad a \diamond x = q, \quad a \diamond r = x, \quad x \diamond q = p, \quad x \diamond p = x, \quad p \diamond a = q,$$

and $r \notin \{x, a, q\}$. Thus $r = p$ or $r \notin \{x, a, q, p\}$. Further,

$$q \diamond s = a, \quad (p \diamond q) \diamond p = s.$$

If $r = p$, then

$$q \diamond a = q, \quad s = (q \diamond q) \diamond q = a \diamond (q \diamond q), \quad s \notin \{x, a, q\},$$

$$H_x(a) = p, \quad H_x(p) = a.$$

If E255 at x fails and $d = 4$, at least one of

$$q \diamond x, \quad p \diamond x, \quad q \diamond q$$

lies outside $\{x, a, q, p\}$.

Proof. Proof certificate. The L_x -orbit identities give

$$x \diamond a = q, \quad x \diamond q = p, \quad x \diamond p = x.$$

Since $q = d_1 = a \diamond x$, we also have $a \diamond x = q$. The recurrence $c_{k-1} = c_k \diamond d_{k+1}$, for $k = 1, 2, 3$, gives

$$a \diamond r = x, \quad q \diamond s = a, \quad p \diamond a = q.$$

We next exclude $r = x, a, q$. If $r = x$, then $a \diamond r = a \diamond x = q$, but $a \diamond r = x$, so $q = x$, impossible on a period-four orbit.

If $r = a$, then (U3) at $k = 0$ gives

$$a = r = x \diamond (q \diamond a).$$

Since $x \diamond x = a$, left cancellation by x gives $q \diamond a = x$. Now (K) with $(y, z) = (q, x)$ gives

$$(q \diamond r) \diamond q = x \diamond ((r \diamond x) \diamond r).$$

With $r = a$, this becomes

$$(q \diamond a) \diamond q = x \diamond ((a \diamond x) \diamond a),$$

hence $p = a$, because the left side is $x \diamond q = p$ and the right side is $x \diamond (q \diamond a) = x \diamond x = a$. This is impossible.

If $r = q$, then $a \diamond q = x$. By (U3) at $k = 1$,

$$(a \diamond q) \diamond a = x \diamond ((q \diamond x) \diamond q),$$

so $q = x \diamond (q \diamond q)$. Comparing with $q = x \diamond a$ gives $q \diamond q = a$. Applying (K) with $(y, z) = (q, x)$ and using $r = q$ gives

$$(q \diamond q) \diamond q = x \diamond (q \diamond q).$$

The left side is $a \diamond q = x$, while the right side is $x \diamond a = q$, another contradiction. Thus $r \notin \{x, a, q\}$, so either $r = p$ or r is outside the principal orbit $\{x, a, q, p\}$.

The identity $q \diamond s = a$ was obtained above, and

$$(p \diamond q) \diamond p = s$$

is the specialization $q \setminus a = (p \diamond q) \diamond p$ from Lemma 9, because $q \diamond s = a$.

Assume now that $r = p$. From (U3) at $k = 0$,

$$p = r = x \diamond (q \diamond a).$$

Since $x \diamond q = p$, left cancellation by x gives

$$q \diamond a = q. \tag{P4.1}$$

Then (U1) at $k = 1$ gives

$$s = a \diamond ((q \diamond a) \diamond q) = a \diamond (q \diamond q). \tag{P4.2}$$

Also (K) with $(y, z) = (q, a)$, together with (P4.1), gives

$$(q \diamond q) \diamond q = a \diamond ((q \diamond a) \diamond q) = a \diamond (q \diamond q) = s. \tag{P4.3}$$

The exclusions for s are immediate. If $s = x$, then $p \diamond x = x$, so Proposition 4 gives $p = q$. If $s = a$, then

$$a = q \diamond s = q \diamond a = q$$

by (P4.1). If $s = q$, then $p \diamond x = q = p \diamond a$, and left cancellation by p gives $x = a$. Hence $s \notin \{x, a, q\}$.

Finally,

$$H_x(a) = ((x \diamond a) \diamond x) = q \diamond x = r = p, \quad H_x(p) = ((x \diamond p) \diamond x) = x \diamond x = a.$$

If E255 at x fails, then $r = q \diamond x \neq x$. The preceding exclusions give either $r \notin \{x, a, q, p\}$, in which case we are done, or $r = p$. In the latter case, if $s = p \diamond x \notin \{x, a, q, p\}$, we are done. The only remaining possibility is $r = p$ and $s = p$. Then (P4.2) and (P4.3) give

$$a \diamond (q \diamond q) = p, \quad (q \diamond q) \diamond q = p. \tag{P4.4}$$

Put $t = q \diamond q$. If $t = x$, then $a \diamond t = a \diamond x = q$, contradicting $a \diamond t = p$. If $t = a$, then (P4.4) gives $a \diamond a = p = a \diamond q$, so left cancellation by a gives $a = q$. If $t = q$, then $t \diamond q = q \diamond q = t = q$, contradicting $t \diamond q = p$. If $t = p$, then $a \diamond t = a \diamond p = x$, contradicting $a \diamond t = p$. Thus $q \diamond q \notin \{x, a, q, p\}$. $\square$

Proposition 11 (Period-five and period-six first data). *If $d = 5$, write*

$$a = c_1, \quad b = c_2, \quad q = c_3, \quad p = c_4, \quad g = q \diamond x.$$

Then

$$\begin{aligned} a \diamond x &= q, & x \diamond q &= p, & x \diamond p &= x, & p \diamond a &= q, \\ g &= a \diamond ((b \diamond a) \diamond b), & q \diamond ((a \diamond q) \diamond a) &= x, & (a \diamond q) \diamond a &= x \diamond (g \diamond q), \end{aligned}$$

and at a bad point $g \notin \{x, b\}$.

If $d = 6$, write

$$a = c_1, \quad b = c_2, \quad c = c_3, \quad q = c_4, \quad p = c_5, \quad h = c \diamond x, \quad g = q \diamond x.$$

Then

$$\begin{aligned} a \diamond x &= q, & x \diamond q &= p, & x \diamond p &= x, & p \diamond a &= q, \\ h &= a \diamond ((b \diamond a) \diamond b), & g &= b \diamond ((c \diamond b) \diamond c), \\ q \diamond ((a \diamond q) \diamond a) &= x, & (a \diamond q) \diamond a &= x \diamond (g \diamond q), \end{aligned}$$

and at a bad point $g \neq x$.

Proof. Proof certificate. Suppose first that $d = 5$. Then $q = c_{d-2} = c_3$ and $p = c_{d-1} = c_4$, so

$$a \diamond x = q, \quad x \diamond q = p, \quad x \diamond p = x.$$

The recurrence at $k = 4$ gives $p \diamond a = q$. By (U1) at $k = 1$,

$$g = d_3 = a \diamond ((b \diamond a) \diamond b).$$

By (U2) at $k = 1$,

$$q \diamond ((a \diamond q) \diamond a) = x,$$

and (U3) at $k = 1$ gives

$$(a \diamond q) \diamond a = x \diamond ((q \diamond x) \diamond q) = x \diamond (g \diamond q).$$

At a bad point, $g = x$ would be $q \diamond x = x$, contrary to the equivalence in Proposition 4; and $g = b$ would make

$$H_x(b) = ((x \diamond b) \diamond x) = q \diamond x = g = b,$$

contrary to Lemma 6.

The $d = 6$ case is the same calculation with $q = c_4$ and $p = c_5$. Thus

$$a \diamond x = q, \quad x \diamond q = p, \quad x \diamond p = x, \quad p \diamond a = q.$$

By (U1) at $k = 1$ and $k = 2$,

$$h = d_3 = a \diamond ((b \diamond a) \diamond b), \quad g = d_4 = b \diamond ((c \diamond b) \diamond c).$$

By (U2) and (U3) at $k = 1$,

$$q \diamond ((a \diamond q) \diamond a) = x, \quad (a \diamond q) \diamond a = x \diamond ((q \diamond x) \diamond q) = x \diamond (g \diamond q).$$

At a bad point $g \neq x$, again by the equivalence in Proposition 4. □

4 Collision and right-tail facts

Proposition 12 (Collision lift). *If $d_i = d_j = e$, then*

$$(c_i \diamond e) \diamond c_i = (c_j \diamond e) \diamond c_j.$$

If also $i \not\equiv j \pmod{d}$, then

$$c_i \diamond e \neq c_j \diamond e.$$

Proof. Proof certificate. (T) gives $x = d_k \diamond ((c_k \diamond d_k) \diamond c_k)$. Equal d_i, d_j allow left cancellation by e . If also $c_i \diamond e = c_j \diamond e = s$, then left-cancel s to get $c_i = c_j$. $\square$

Proposition 13 (Right-collision rectangles). *If $y \diamond x = r$, then*

$$(y \diamond r) \diamond y = r \setminus x. \tag{R}$$

Consequently, if $y \diamond x = z \diamond x = r$ and $y \diamond r = z \diamond r$, then $y = z$.

Proof. Proof certificate. (T) with $(y, z) = (y, x)$ gives $x = r \diamond ((y \diamond r) \diamond y)$. This is (R). The last claim is left cancellation after applying (R) to y and z . $\square$

Lemma 14 (Right-tail first repetition). *Assume $q \diamond x \neq x$. Define*

$$\rho_0 = x, \quad \rho_{n+1} = \rho_n \diamond x.$$

Then $\rho_n \neq x$ for $n \geq 1$. Let $1 \leq i < j$ be the first repetition in the tail, with j minimal, and put

$$e = \rho_i = \rho_j, \quad y = \rho_{i-1}, \quad z = \rho_{j-1}.$$

Then

$$\begin{aligned} y &\neq z, & y \diamond x &= z \diamond x = e, \\ (y \diamond e) \diamond y &= (z \diamond e) \diamond z = e \setminus x, & y \diamond e &\neq z \diamond e. \end{aligned}$$

Proof. Proof certificate. If some $\rho_n = x$, then $\rho_{n-1} \diamond x = x$, forcing $\rho_{n-1} = q$, contradiction. Finiteness gives a first repetition. Minimality gives $y \neq z$. Proposition 13 gives the rectangle and splitter. $\square$

Proposition 15 (Left-translation labels). *Write $a \diamond b = \sigma(a)(b)$. Then $\sigma(a) \in \text{Sym}(M)$ and the label map $a \mapsto \sigma(a)$ is injective. If $y \diamond x = z \diamond x = r$ with $y \neq z$, then*

$$\begin{aligned} \sigma(y)(x) &= \sigma(z)(x) = r, & \sigma(y)(r) &\neq \sigma(z)(r), \\ \sigma(\sigma(y)(r))(y) &= r \setminus x = \sigma(\sigma(z)(r))(z). \end{aligned}$$

Proof. Proof certificate. Left translations are permutations by Proposition 1. Suppose $L_y = L_z$. Choose any $x \in M$ and put

$$e = y \diamond x = z \diamond x.$$

The two E677 specializations give

$$x = y \diamond (x \diamond (e \diamond y)), \quad x = z \diamond (x \diamond (e \diamond z)).$$

Since $L_y = L_z$, the second equality may be written as

$$x = y \diamond (x \diamond (e \diamond z)).$$

Comparing with the first equality and left-cancelling successively by y , then by x , then by e , gives $y = z$. Thus $a \mapsto \sigma(a)$ is injective.

If $y \diamond x = z \diamond x = r$ and $y \neq z$, Proposition 13 gives

$$(y \diamond r) \diamond y = (z \diamond r) \diamond z = r \setminus x.$$

It also says that $y \diamond r = z \diamond r$ would force $y = z$, so $\sigma(y)(r) \neq \sigma(z)(r)$. Rewriting these products in the notation $a \diamond b = \sigma(a)(b)$ gives the displayed formulas. $\square$

5 Bad-point checklist and open gates

Proposition 16 (Bad-point package). *If E255 at x fails, then:*

1. the L_x -orbit of x has period $d \geq 4$;
2. no u satisfies $u \diamond x = x$, so $x \notin \text{im } R_x$;
3. R_x has a canonical nontrivial collision fiber from Lemma 14;
4. on a fiber $R_x^{-1}(e)$ as above, $t \mapsto t \diamond e$ is injective and obeys (R);
5. H_x and F_x are fixed-point-free;
6. if $d = 4$, some one of $q \diamond x, p \diamond x, q \diamond q$ is outside $\{x, a, q, p\}$.

Proof. Proof certificate. Items: Corollary 8; Proposition 4; Lemma 14; Proposition 13; Lemma 6; Proposition 10. $\square$

Remark 17 (Dependency table for a bad point). The following implications are the current compressed route graph. They are all proved above.

$$\begin{aligned} \text{E255 at } x &\iff q \diamond x = x \\ &\iff \exists u (u \diamond x = x) \\ &\iff q \setminus x = x \\ &\iff ((x \diamond x) \diamond q) \diamond (x \diamond x) = x \\ &\iff (q \diamond x) \diamond q = p. \end{aligned}$$

Under the negation $q \diamond x \neq x$:

$$\begin{aligned} \text{no } u \diamond x = x &\implies x \notin \text{im } R_x \\ &\implies R_x \text{ has a nontrivial fiber} \\ &\implies y \diamond x = z \diamond x = e, \ y \neq z \\ &\implies (y \diamond e) \diamond y = (z \diamond e) \diamond z = e \setminus x, \ y \diamond e \neq z \diamond e. \end{aligned}$$

The canonical way to choose the fiber is the first repetition in the right tail $\rho_{n+1} = \rho_n \diamond x$. This removes choice from the splitter route.

For the orbit route:

$$\begin{aligned}
q \diamond x \neq x &\implies d \geq 4, \\
d = 4 &\implies q \diamond x \in \{p\} \cup (M \setminus \{x, a, q, p\}), \\
d = 4, q \diamond x = p &\implies p \diamond x \notin \{x, a, q\}, \\
d = 4, q \diamond x = p, p \diamond x = p &\implies q \diamond q \notin \{x, a, q, p\}.
\end{aligned}$$

So a bad period-four point always creates an external product. No four-element principal-orbit-only bad point survives these facts.

Remark 18 (Finite facts used, nothing else). Only these finite principles appear: finite surjection = bijection; a finite self-map missing a point has a nontrivial fiber; every finite forward orbit repeats; and conjugate finite self-maps have the same cycle structure. Every cancellation in the paper is left cancellation by a proved bijective left translation.

Remark 19 (Do not use).

No right cancellation. No associativity. No identity element.
No universal idempotence. No global “ H_x has no cycles”.
Do not use $d_k \setminus x = c_{k-2} \diamond c_{k-1}$.
Do not use $p \diamond d_k = d_{k+1}$ unless freshly proved.

Gate 20 (Fixed-point gate). Prove $q \diamond x = x$, equivalently Lemma L or $(q \diamond x) \diamond q = p$, with a visible finite step.

Gate 21 (Splitter gate). Use the canonical right-tail collision fiber, or any collision $d_i = d_j = e$, to turn the injective splitter $t \mapsto t \diamond e$ into either $q \diamond x = x$ or a finite counting contradiction.

Gate 22 (Period-four external gate). At a bad period-four point, feed the forced external element among

$$q \diamond x, \quad p \diamond x, \quad q \diamond q$$

back into the fixed-point gate or the splitter gate.

References

- [1] M. Bolan et al. The Equational Theories Project: Advancing collaborative mathematical research at scale. arXiv:2512.07087, 2025.
- [2] Equational Theories Project. Dashboard. https://teorth.github.io/equational_theories/dashboard/.
- [3] Equational Theories Project. Blueprint, Chapter 13: Equation 677. https://teorth.github.io/equational_theories/blueprint/677-chapter.html.

- [4] Equational Theories Project. Blueprint, Chapter 5: Finite implications and infinite models. https://teorth.github.io/equational_theories/blueprint/infinite-model-chapter.html.